Module 2 Practicum

By: Andy Steiner

Section 1: Brain Area Report - Dorsolateral Prefrontal Cortex (dlPFC)

Subsection 0: In section 1, I will investigate ECoG data for the surface electrode dorsolateral prefrontal cortex surface electrode (RAF). The structure and function of the brain area will be discussed, as well as the predominant cell type and cognitive networks in which it is involved. The dlPFC plays a major role in executive functioning and overall cognition. It is known to facilitate planning, attention, and decision-making. By observing trends in ECoG data, I will demonstrate the dlPFC's role in cognition.

Subsection 1: Dorsolateral Prefrontal Cortex (dlPFC)

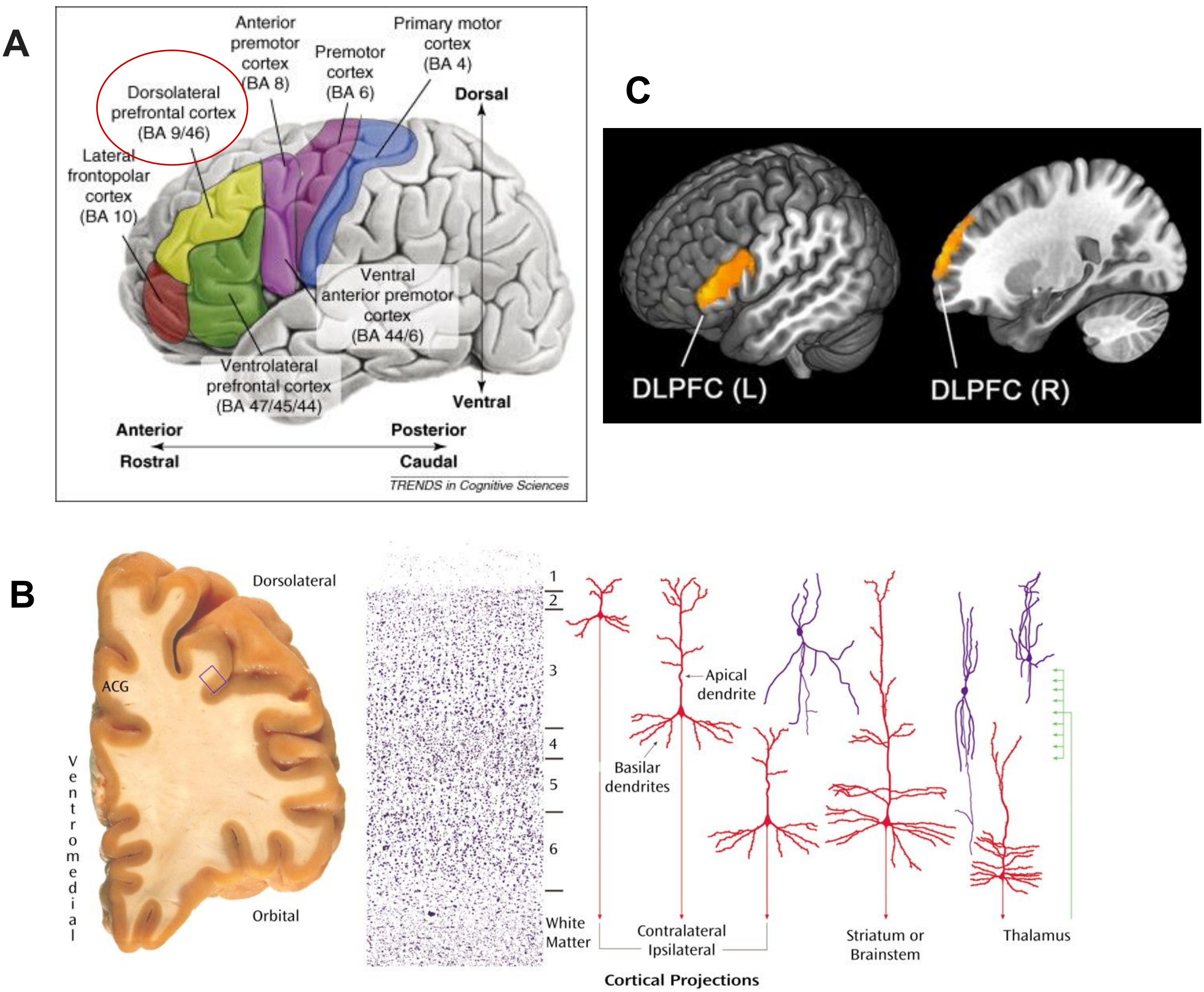

Figure 1. The dorsolateral prefrontal cortex is a subsection of the prefrontal cortex involved in many executive and cognitive functions including emotional regulation, memory, and decision making.

(A) The dlPFC is a bilateral structure in the frontal lobe of the brain. It is part of the middle frontal gyrus, located anterior to the premotor cortex, and encompasses the lateral section of Brodmann areas 9 and 46. However, due to individual variability, defining the dlPFC solely by anatomical landmarks can be challenging. For this reason it is often functionally defined rather than structural-anatomically (Hertrich et al., 2021). **(B)** The dlPFC has a typical six-layer neocortical structure. Layers 2 and 3 are crucial for intracortical communication,layers 5 and 6 for subcortical outputs. This allows the dlPFC to send and receive information from almost all cortical sensory systems, motor systems, and many subcortical structures. (Miller & Cohen, 2001). The predominant cell type comprising of about 75% of the dlPFC is the excitatory pyramidal neuron (red cells). The remaining 25% are primarily local inhibitory interneurons (purple cells). **(C)** Shown is FMRI data where baseline scans are subtracted from premeal and satiated states. The study found increased BOLD signal in the dlPFC when satiated. It is hypothesized that the reduced motivation to eat associated with satiation is partly mediated by prefrontal brain regions (Thomas et al., 2015).

Subsection 2: Functions of the Dorsolateral Prefrontal Cortex (dlPFC)

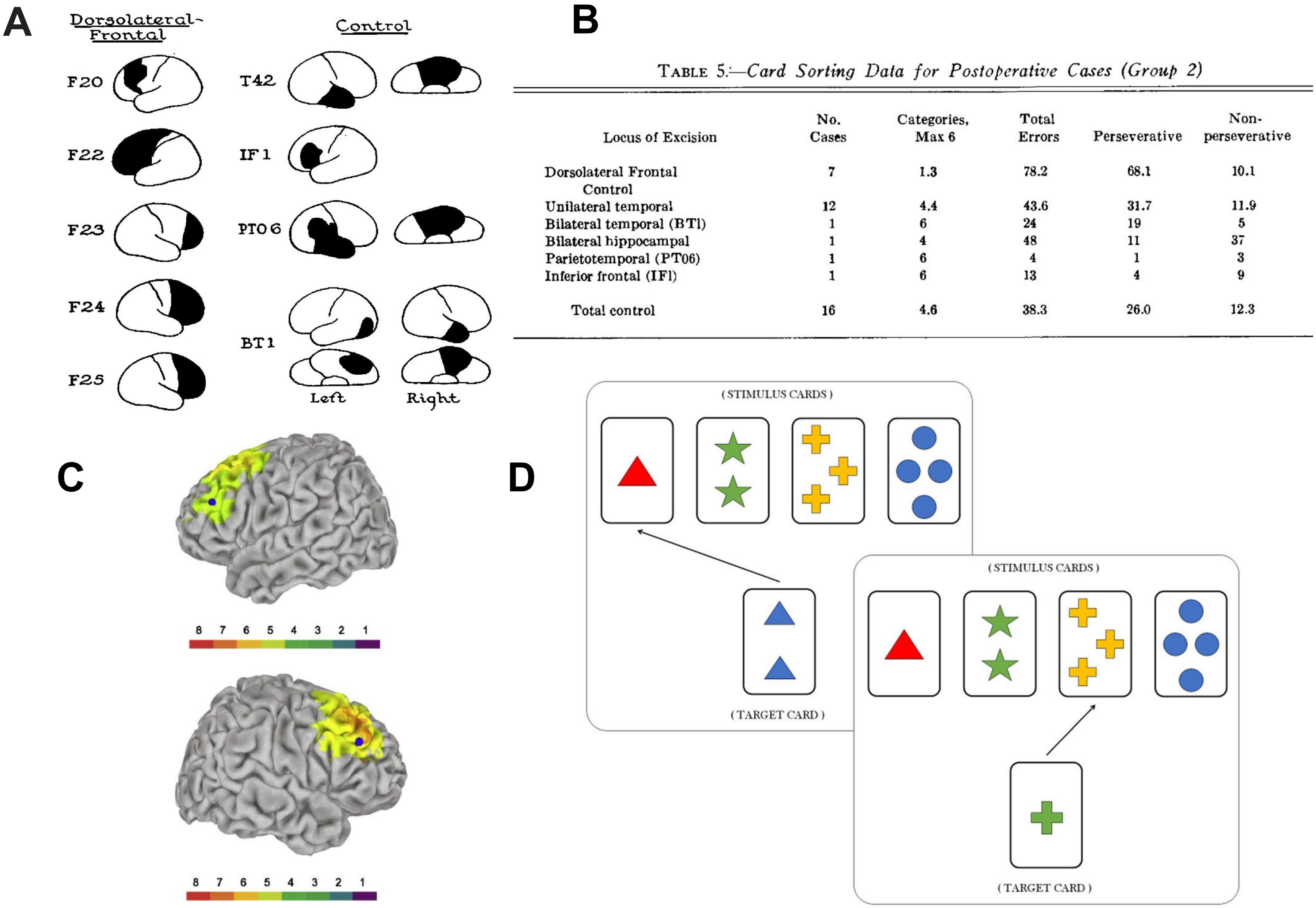

TABLE 5.—*Card Sorting Data for Postoperative Cases (Group 2)*

Locus of Excision	No. Cases	Categories, Max 6	Total Errors	Perseverative	Non-perseverative
Dorsolateral Frontal	7	1.3	78.2	68.1	10.1
Control					
Unilateral temporal	12	4.4	43.6	31.7	11.9
Bilateral temporal (BT1)	1	6	24	19	5
Bilateral hippocampal	1	4	48	11	37
Parietotemporal (PT06)	1	6	4	1	3
Inferior frontal (IF1)	1	6	13	4	9
Total control	16	4.6	38.3	26.0	12.3

Figure 2. The dlPFC plays a role in many higher-order cognitive processes. Research suggests it plays a critical role in working memory.

(A) A study (n=91) investigated differences in working memory performance among those with specific lesions and/or lobectomies during the Wisconsin Card Sorting Test. The test involved picking up on a rule change in the sorting game (color, number, color), that changed without warning. To succeed in the task, participants needed to abandon the current method of approach for a new method. Sticking with an old method, after the rule was shifted is classified as a "perseverative error". All other errors are non perseverative. **(B)** Data for group 2 (n=23) is shown. Despite age and postoperative IQ scores remaining constant among groups, subjects with specifically dorsolateral frontal lobectomies had the most errors during the memory task, and significantly higher perseverative errors than control groups (68.1% vs 26.0%). Despite this study being well known, its small sample size and varying types of trauma are certainly limitations (Brenda, 1963). **(C)** Instead of relying on broad lesion locations, a more recent study found similar working memory differences by analyzing CT scans (shown), and correlating the location of the specific brain damage to scores on working memory tests. The authors state that working memory is supported by a network of left-lateralized brain areas, and that specifically the left dlPFC is critically important to support working memory (Barbey et al., 2013). **(D)** An example of the Wisconsin Card Sorting Test (WCST) is shown. This illustrates how different rules (shape, number, color) could be used to sort the cards. In this example, shape is the rule (Alessandro et al., 2020). Damage to the dlPFC, from stroke, trauma, or disease leads to dysexecutive syndrome. This condition is characterized by a complex breakdown of cognitive management. Patients have difficulty planning, initiation/stopping actions, and adapting to new rules/events. A key symptom is perseveration with an old rule despite negative feedback, which why the WCST is often used to assess the integrity of the dlPFC.

Subsection 3: ECoG data for the Dorsolateral Prefrontal Cortex (RAF)

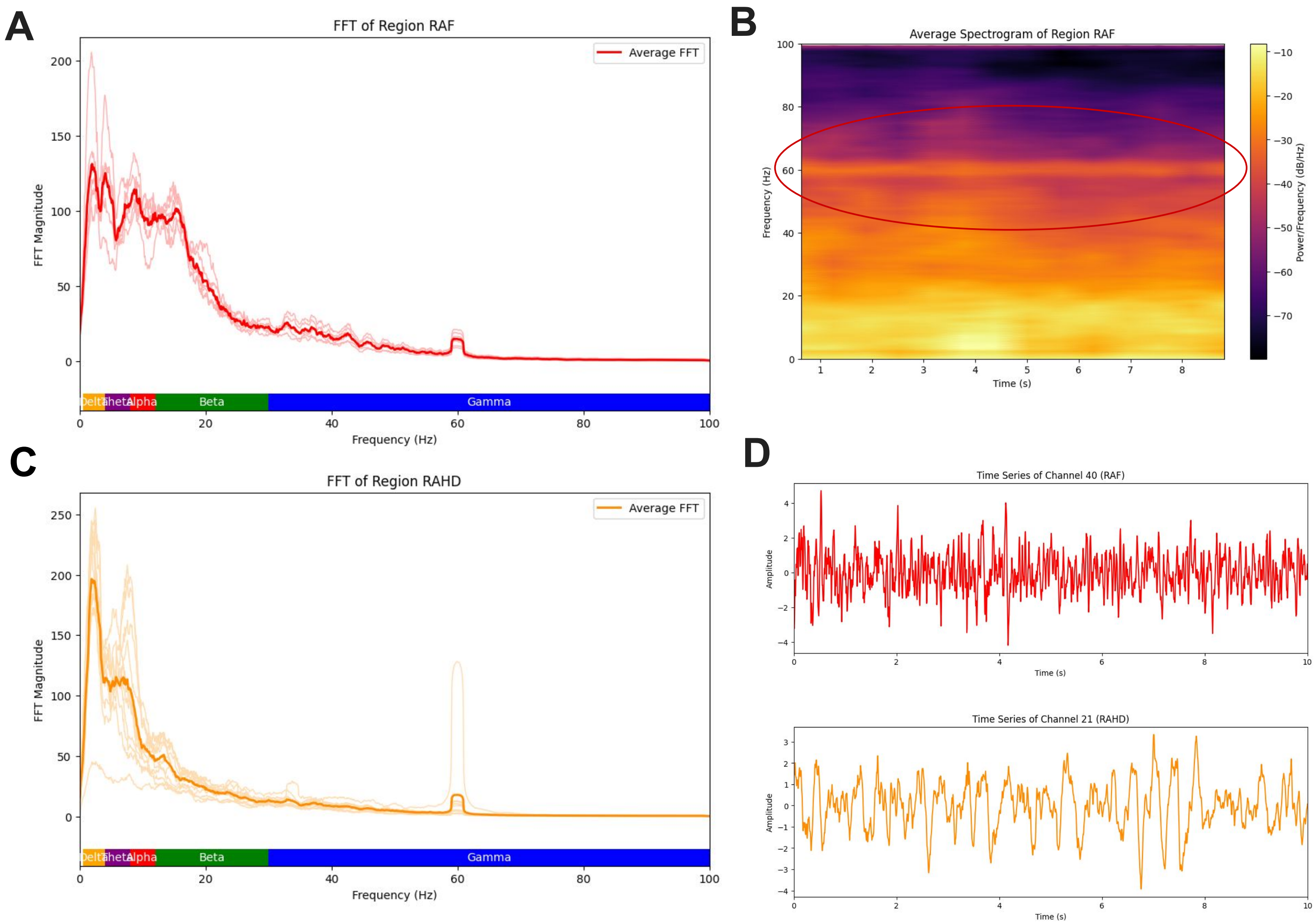

Figure 3. Electrocorticography (ECoG) data for the RAF, and RAHD regions are compared. The RAF region contains the dlPFC, and the RAHD region passes through the hippocampus.

(A) Shown is the Fast Fourier Transform (FFT) for the RAF region. The region appears to be dominated by low frequencies between 0.5-20 Hz, with a noticeable decrease in theta rhythm activity (4-8 Hz). Due to this region being a surface electrode, it is likely picking up a mix of other components when compared to depth electrodes. **(B)** The average spectrogram of the RAF region is shown. The graph shows higher activity in lower frequencies, decreasing after about 30 Hz. There is a clear band around 60 Hz due to electrical grid noise (circled red). This peak at 60 Hz can be seen in most graphs and should be ignored. **(C)** The FFT for the RAHD region is shown for comparison. This graph clearly shows a higher magnitude of lower frequencies when compared to the RAF FFT graph. Specifically, more delta and theta activity between 0.5-8 Hz. This matches with our understanding of the hippocampus, as it is known to produce strong slow oscillations. Literature suggests that these slower oscillations, especially delta waves are crucial for memory consolidation (Kim et al., 2020). **(D)** The raw ECoG data for the RAF (top) and RAHD (bottom) are shown. It is clear even just by eye that the RAHD graph is much more spaced out than the RAF graph. This shows the powerful low-frequency delta and theta rhythms in the hippocampus crucial for memory processes, compared to the higher frequencies present in the RAF which are linked to active concentration, and logical thinking.

Section 2: ECoG oscillation report - Gamma

Subsection 0: In section 2, I will analyse gamma oscillations. Using scientific literature and ECoG data, I will show which brain areas exhibit the most gamma oscillations, their significance, as well as the neurocircuitry responsible. Gamma oscillations are linked to high cognitive function and attention, specifically cognitive processes including information processing, memory, and perception. Studies have linked abnormal gamma oscillations to conditions of the central nervous system, including Parkinson's disease, Alzheimer's disease, and schizophrenia. I will also describe differences in frontal lobe EEG vs ECoG, and how to recognise and eliminate noise from EEG data.

Subsection 1: Gamma Oscillations

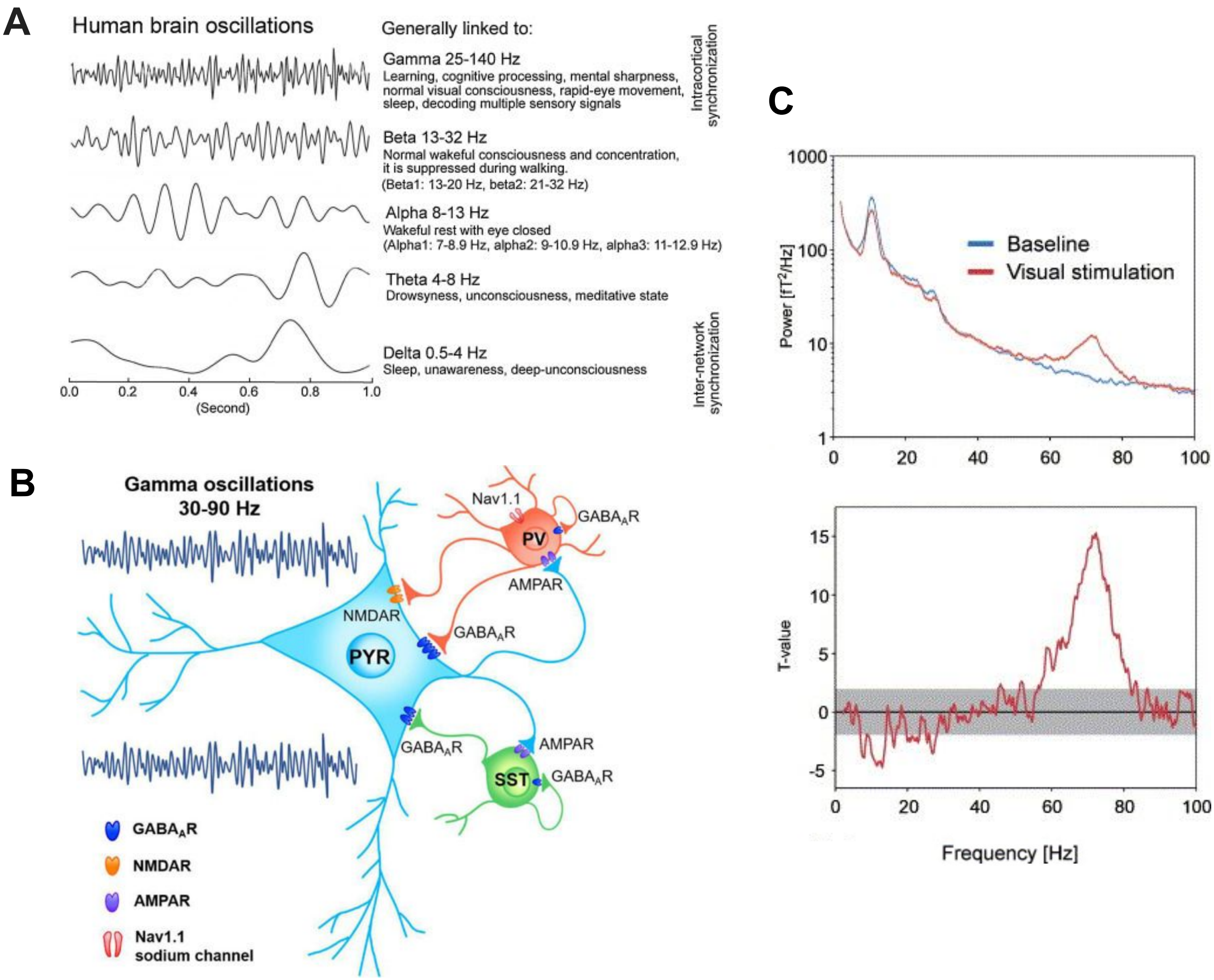

Figure 1. Gamma oscillations are a class of fast brain waves >25 Hz that reflect localized and highly engaged neural computation.

(A) Gamma oscillations comprise the largest frequencies range in the brain (typically ~25-140 Hz), and are the fastest brain waves. Gamma rhythms are closely linked to specific sensory inputs (e.g., seeing a specific image) or internal cognitive processes (e.g., holding an item in working memory). They are thought to be a crucial part of combining together different features of a stimulus, and specifically controlling the connectivity between different brain regions (Jafari et al., 2020). **(B)** The generation of gamma oscillations is most commonly explained by the Pyramidal-Interneuron Network Gamma (PING) model. In this circuit, an excitatory pyramidal cell fires, which in turn excites a fast-spiking inhibitory interneuron. This interneuron then sends a wave of inhibition back to a group of pyramidal cells, temporarily shutting them down. About ~15-20 milliseconds later, the pyramidal cells fire again restarting the cycle. This feedback loop of excitation-inhibition-release creates fast rhythmic oscillations in the gamma frequency range (Guan et al., 2022). **(C)** An average power spectrum (logarithmic axis) during baseline and a visual stimulation task (top), and the t values for the difference between the power during baselines and visual stimulation (bottom) is shown. The task involved pressing a button when participants detected a change in a moving stimuli. The data shows a significant increase in gamma oscillations (60-80 Hz) during the task, compared to other frequencies. This supports the idea that gamma rhythms are important for attention-demanding tasks, as well as recognising differences in moving stimuli (Hoogenboom et al., 2006).

Subsection 2: Mapping Gamma Oscillations

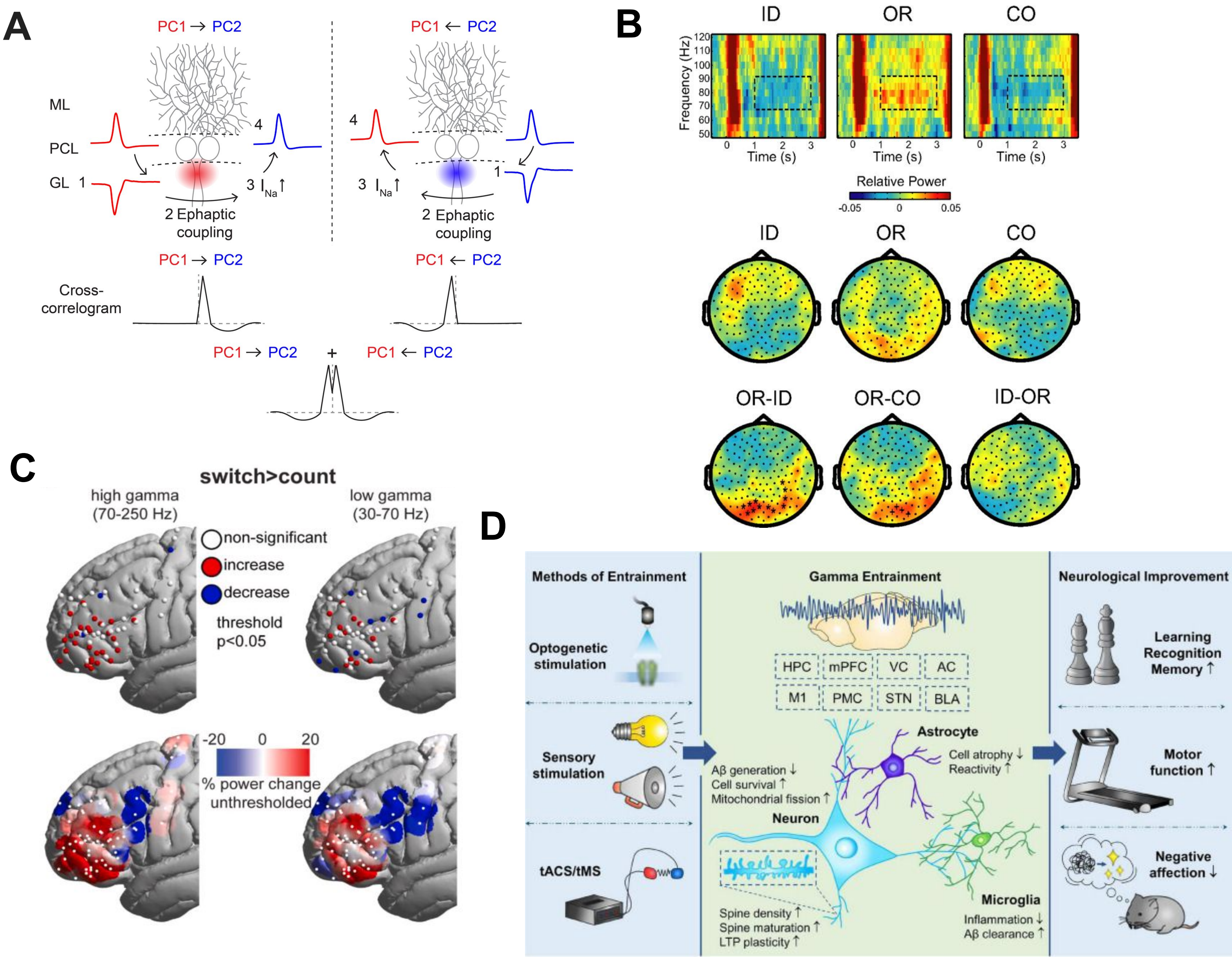

Figure 2. Gamma oscillations are most strongly exhibited in task-engaged sensory and cognitive hubs like the occipital and frontal lobes, and their dysfunction is a key therapeutic target in neuropsychiatric disease.

(A) Gamma oscillations typically occur in higher functioning grey matter areas that can support the PING architecture. Interestingly, certain axons can fire in rapid bursts that fall within the gamma range without using PING circuitry. It is thought that ephaptic coupling (shown) as well as gap junctions contribute to axon-axon interactions that can lead to rapid oscillations present in the gamma range (Han et al., 2020). **(B)** A study investigated brain activity during a visuospatial delayed-match-to-sample (DMS) task using magnetoencephalography (MEG). Higher gamma activity was observed in the occipital lobe during the orientation task (OR), despite the identity task being harder. Researchers hypothesize that gamma activity for visual processing plays a larger role in the dorsal stream (where/how) rather than ventral stream engagement (what). This shows that gamma activity in the occipital lobe is specifically involved in maintaining spatial working memory representations. ID-CO difference is omitted due to lack of significant difference (Jokisch et al., 2007).

(C) ECoG data was collected during a task where participants were asked to count aloud 1-20 (count), and compared it to a more cognitive demanding "switch" task where patients had to alternate between letters and numbers (e.g 1, a, 2, c). Shown is the "switch > count" contrast. Patients had clear increases in high gamma and beta oscillations during the working memory task (Assem et al., 2023). **(D)** Abnormal gamma oscillations in patients is linked to conditions of the central nervous system, including Alzheimer's disease, Parkinson's disease, and schizophrenia. A technique called gamma entrainment using sensory stimuli (GENUS) can be a potential therapeutic approach for many neuropsychiatric diseases, and can provide significant neuroprotection. Methods of Entrainment include optogenetic stimulation, sensory stimulation of the auditory and visual cortex, as well as transcranial magnetic stimulation (tMS) and transcranial alternating current stimulation (tACS). While further research is needed, these techniques could be non-invasive and cost effective treatments of CNS diseases (Guan et al., 2022).

Subsection 3: ECoG vs EEG data

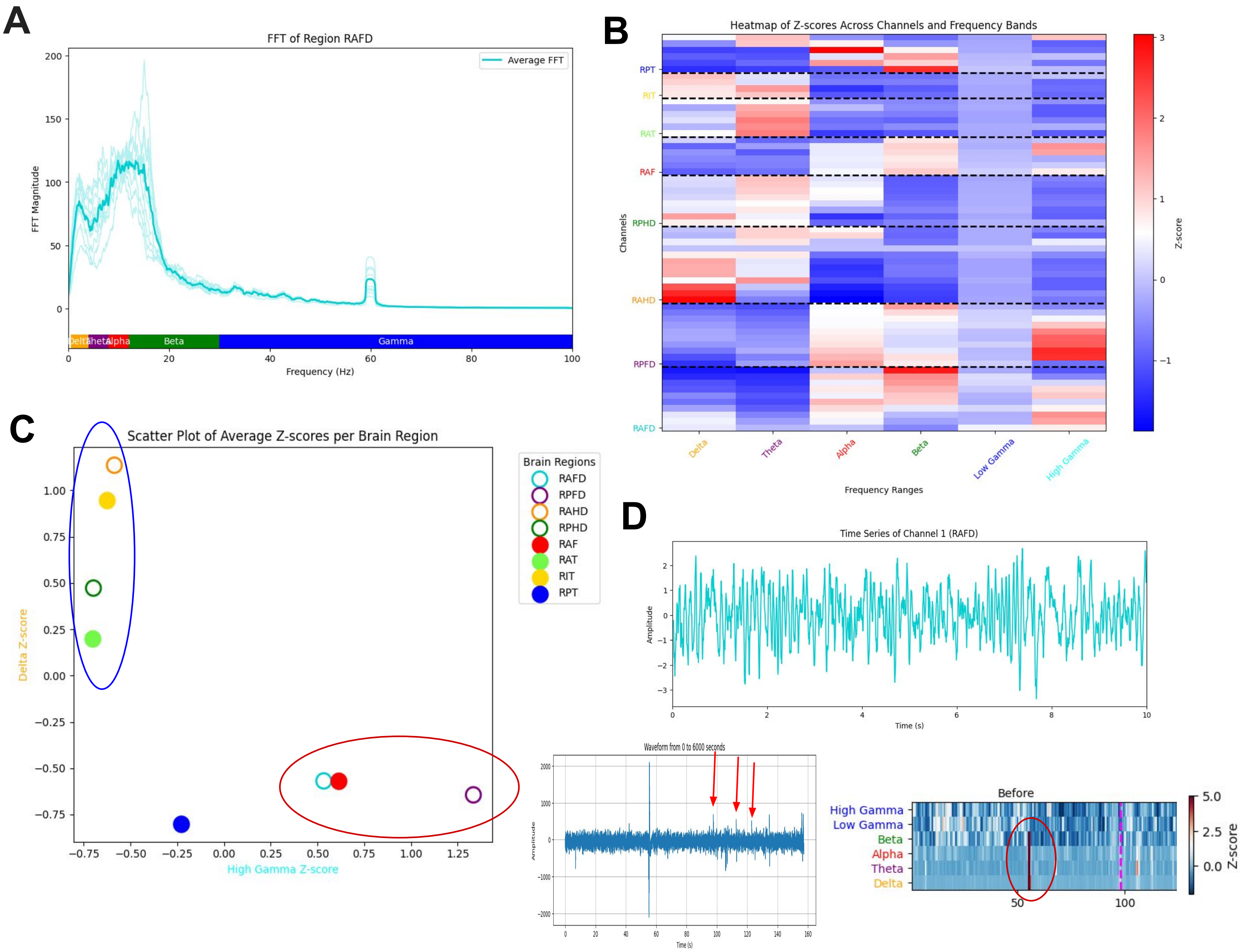

Figure 3. Frontal lobe regions exhibit a specialization for high-frequency gamma oscillations over low-frequency delta rhythms, and EEG scalp recordings are more susceptible to noise than intracranial ECoG data.

(A) The Fast Fourier Transform (FFT) for the prefrontal cortex depth electrode (RAFD) is shown. This region, as well as the RAF region discussed in **Section 1**, are closest in location to the EEG recording site used in **Section 3**. The RAFD and RAF regions have elevated high gamma, alpha, and beta levels. While lower frequency oscillations such as delta and theta waves are less prominent compared to other regions. **(B)** Average z-scores across all channels and frequency bands is shown. This figure demonstrates a trend that region with high frequency oscillations typically have less low frequency activity, and vice versa. High frequency waveforms dominate frontal lobe regions while slower waves are more present in the hippocampal regions (see RAFD/RPFD vs RAHD). **(C)** The scatterplot of average z-scores for all brain regions showing high gamma activity of the x-axis and delta activity on the y-axis is shown. The scatter plot further supports the idea regions typically support either high or low frequency activity. It also shows the grouping of frontal lobe regions (circled red), compared to more temporal regions that express lower frequencies. **(D)** The raw ECoG data (top) is compared to raw EEG data (left) and z-scored spectrogram for the EEG data (right). The ECoG data is much cleaner and due to them being implanted within the skull, and do not pick up on certain eye movements or other noise. Both graphs will still show electrical grid noise at around 60 Hz. The red arrows point to blink noise that is present in the EEG data, and the circled bar on the spectrogram is likely cell phone noise or large movements due to it being present across most frequencies. During the task is **Section 3**, participants were instructed to move as little as possible, and put their phones in airplane mode. Trails with noticeable cell phone or other noise, were retaken. The spikerbox additionally filters certain high and low frequency activity that would be out of the range for typical brain activity.

Section 3: EEG experimental data

Subsection 0: In section 3, I will perform an experiment where subjects do a mental math task while frontal lobe EEG data is recorded. Participants will perform multi-step linear equations on a computer. They must do these calculations in their head, which requires holding numbers in working memory, as well as simple arithmetic. Due to the attention and concentration required to do mental arithmetic, it is hypothesized that we will see clear increases in theta and gamma oscillations during the task compared to rest periods. However, due to noise, we might not get clear theta readings from the EEG. There are a total of 5 participants who did the task twice (2 trials), for a total of 10 trials. There is a period of rest for 1 minute and 30 seconds before and after the task, and the task was performed for about 2.5 minutes.

Subsection 1: Methods for EEG experiment

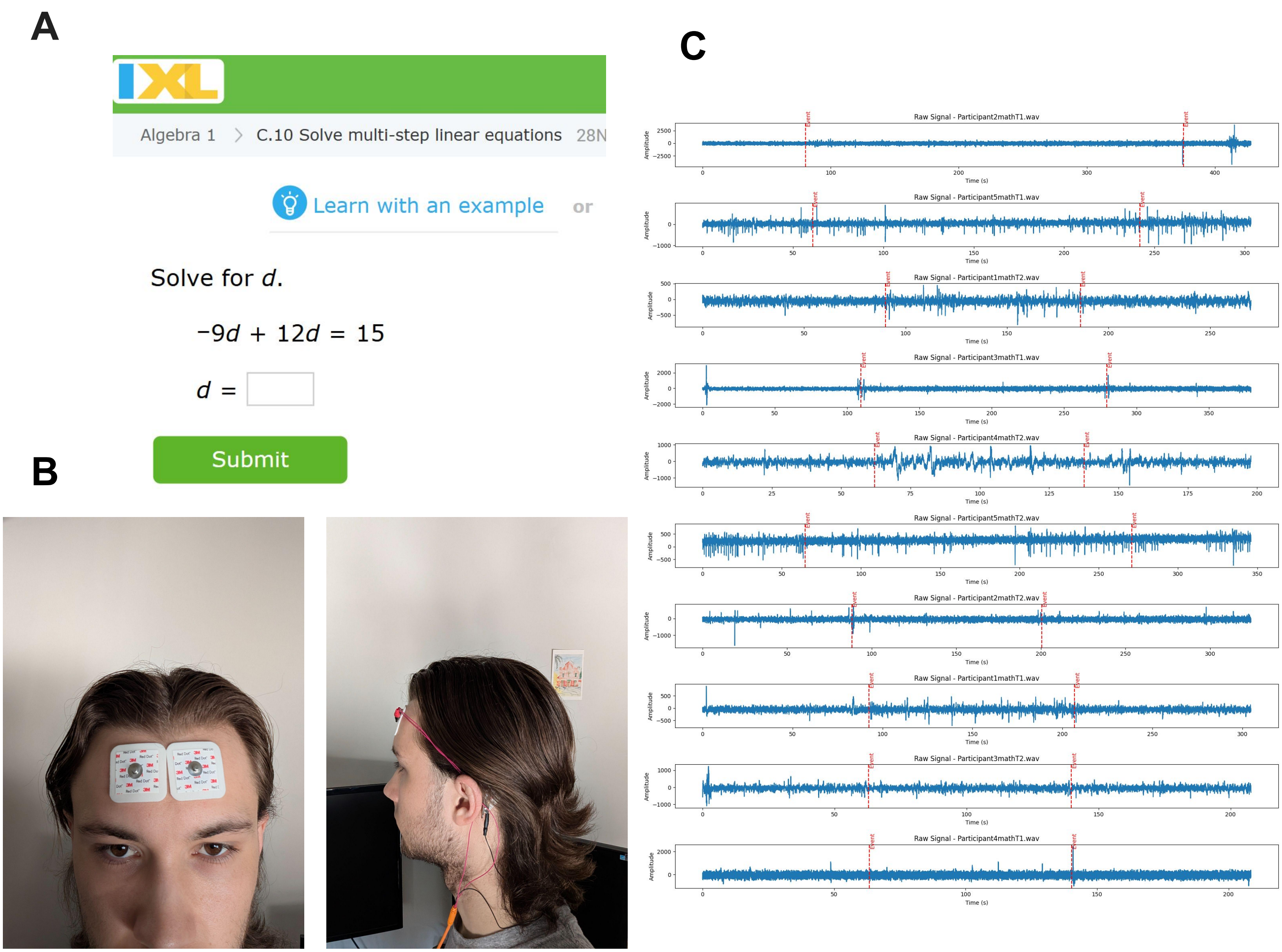

Figure 1. Experimental methods for collecting frontal lobe EEG activity during a math task.

(A) Shown is an example of the math task. The task was specifically to solve multi-step linear equations mentally. Participants were instructed to try and get as many correct as quickly as possible to ensure cognitive load, however times and scores were not recorded. Participants began the trial with a 1 minute 30 second period of open eye rest (staring at fixed point on wall). The task was performed for about 2.5 minutes, followed by another open-eyed rest period for 1.5 minutes. To reduce noise, participants were instructed to move as little as possible, not bump cables, and put their phone in airplane mode. The computer being used was also run on battery to eliminate charging noise. A total of 10 trials from 5 participants were collected. **(B)** Electrode placement during the task is shown. Two electrodes were placed close together on the forehead, and the grounding electrode was placed behind the ear on the mastoid bone. We aim to record from the frontal cortex, specifically prefrontal regions. **(C)** Raw EEG data for the 10 trials is shown. All participants were between the ages of 20-22. Subjects 1-5 disclosed having at least one caffeinated beverage, and subject 2 additionally reported taking Adderall XR 10mg for ADHD management. Subject 7 was the only left handed participant. All participants were male. There were no other relevant medications or conditions to report.

Subsection 2: EEG data processing

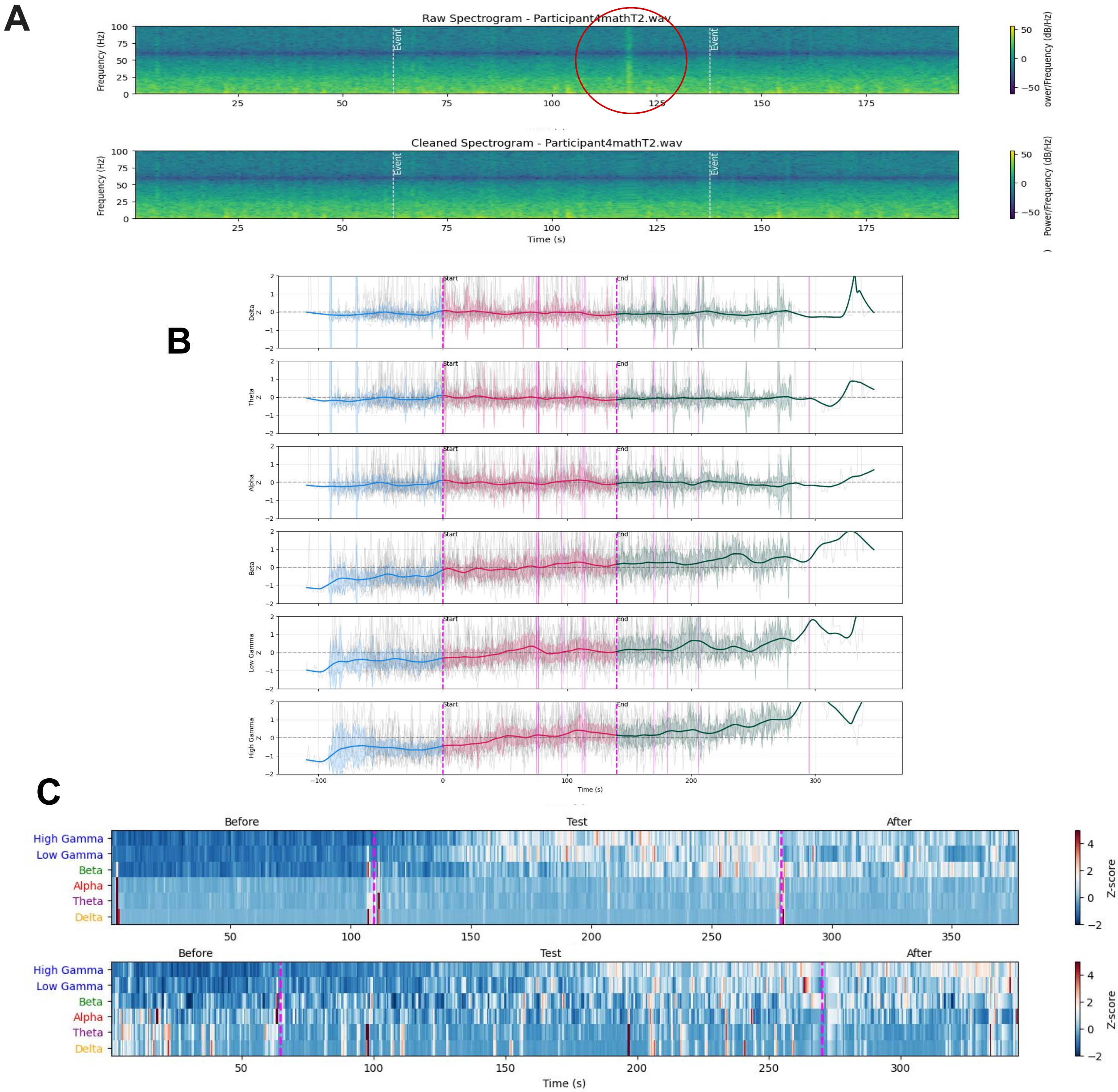

Figure 2. EEG data was processed using google colab code, and noise is removed using cleaned spectrograms.

(A) The raw (top) vs cleaned (bottom) spectrogram for the EEG data is shown. Noise that spans across all regions, such as cell phone noise, movement, and eye blinks, is removed from the raw data (circled red). This is done for all the trials before further analysis in conducted. **(B)** Average power band curves for each frequency is shown. This step shows how the raw eeg data is separated into different frequencies. In our experiment, there appears to be small increases in high gamma throughout the task, which remains elevated even after the task has completed. **(C)** The processed data for two trials is shown. This data averaged across all trials is what creates the diagrams shown in (B). This diagram clearly shows low high/low gamma activity before the task, highest gamma activity during the task, then elevated levels after the task has ended. We hypothesize that 1.5 minutes might not be enough time to “reset” the mind back to baseline, and a longer period of rest might yield oscillations closer to baseline. Individual differences in participants could be due to some people finding the problems easier than others, differences in caffeine/substance intake, or differences in level of concentration.

Subsection 3: EEG Results

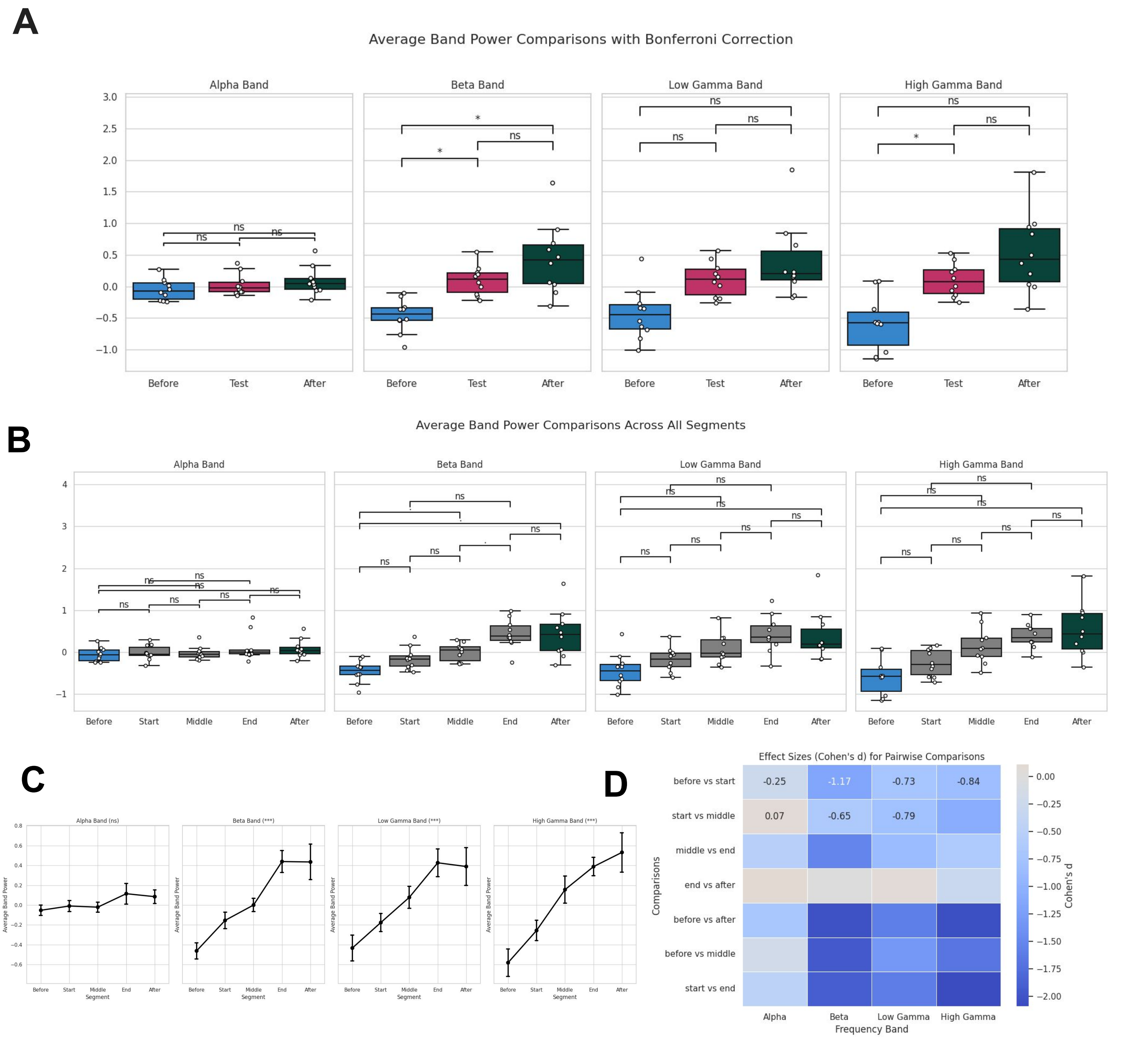

Figure 2. EEG data did not show any significant differences, although there appears to be increases in gamma activity following the test, more trials are needed with fewer pairwise comparisons to investigate any effects.

(A) Average power bands for the different frequencies during before, test, and after are shown. The trend shows an increase in higher frequency activity as the task progresses, although no significant differences between groups were found. **(B)** Average power band curves across all segments is shown. This graph separates the start, middle, and end of the trial. This graph is more specific and you can better see trends in high frequency oscillations. It appears that the rise in activity in beta, low gamma, and high gamma seems to decrease once the test ends, but there are still no significant differences between any groups. We hypothesize that longer post-trial periods or more subject could be needed to yield significant results. **(C)** Average power band across segments graph is shown. This also illustrates that alpha and gamma activity increased throughout the task, and you can see the slope of the line decrease during the “after” period for beta and low gamma. **(D)** Shown is the effect size (cohen’s d for all pairwise comparisons). Largest effect sizes were seen between before vs after, before vs middle, and start vs end periods for beta/gamma oscillations, We expect that results might not be significant due to bonferroni corrections of multiple pairwise corrections. If our original hypothesis of start vs middle and middle vs end might yield significant results if fewer pairwise comparisons are conducted.

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